

Comparison of boundary conditions to describe drying of turmeric (*Curcuma longa*) rhizomes using diffusion models

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Abstract Turmeric is harvested with high moisture content and should be dried before the storage. It is observed that drying is quickest when the rhizomes are peeled and cut in small cylindrical pieces. In order to describe the process, normally a diffusive model is used, considering boundary condition of the first kind for the diffusion equation. This article uses analytical solutions considering boundaries conditions of the first (model 1) and third (model 2) kinds coupled to an optimizer to describe the drying process. It is shown that, for model 1, the fit of the analytical solution to the experimental data is biased, despite the good statistical indicators (chi-square χ^2 equal to 1.7095×10^{-3} and coefficient of correlation R^2 of 0.9988). For model 2, the errors of the experimental points about the simulated curve can be considered randomly distributed, and the statistical indicators are much better than those obtained for model 1: $\chi^2 = 3.5596 \times 10^{-4}$ and $R^2 = 0.9996$.

Keywords Analytical solution · Effective mass diffusivity · Convective mass transfer coefficient · Cauchy boundary condition · Finite cylinder

Introduction

Curcuma longa is a native plant from India used as spice and also to obtain extracts. Due to the importance of its components as anti-oxidant, anti-inflammatory, anti-mutagenic, anti-carcinogenic and anti-microbial, among others, several studies about this agricultural product are found in the literature (Negi et al. 1999; Kim et al. 2000; Quiles et al. 2002; Surh 2002; Inano and Onoda 2002; Mohanty et al. 2004; Prasad et al. 2006; Xia et al. 2007; Adaramoyea et al. 2009; Chen et al. 2010; Yue et al. 2010). In general, when the product is harvested, the rhizomes contain high moisture content. According Silva et al. (2007), water is a constituent part present in high concentration in fresh food. It influences considerably the taste, the process of digestion and the physical structure of agricultural products. Practically all of the deterioration processes taking place in food are influenced in some way or other by the concentration and mobility of water in its interior. So, after the harvest and before the storage or transportation, in general, the rhizomes are submitted to the drying process. Some works describing drying of turmeric are found in the literature. For instance, Prasad (2009) described drying of *Curcuma longa* in open sun, and the thermal behavior of the process was investigated. In this research, the effect of the difference between the product temperature (at the surface) and the temperature surrounding on the convective heat transfer coefficient was studied. On the other hand, Prathapan et al. (2009) investigated the effect of the heat treatment on curcuminoid; color value and total polyphenols of fresh turmeric rhizome. According Devahastin and Niamnuy (2010), not only nutritional changes occur during drying but also other ones including physical and microstructural changes which are of importance and need to be optimized, preferably through the use of various modeling

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使用扩散模型来比较描述姜黄根茎干燥过程的边界条件

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摘要: 姜黄收获时其含水量较高, 因此在储存之前需要进行干燥处理。研究发现, 当将根茎剥皮并切成小圆柱形块时, 干燥过程最为迅速。为了描述这一过程, 通常采用扩散模型, 并考虑第一类边界条件对扩散方程的影响。本文提出了解析解, 结合第一类 (模型 1) 和第三类 (模型 2) 边界条件, 通过优化算法来描述干燥过程。结果表明, 对于模型 1, 尽管统计指标良好 (卡方值等于 1.7095×10^{-0} , 相关系数 R 为 0.9988), 但解析解与实验数据的拟合程度存在偏差。而对于模型 2, 实验点围绕模拟曲线的误差可以被认为是随机分布的, 其统计指标明显优于模型 1 的结果。5596 $\times 10$, $R00.9996$ 。

关键词: 解析解、有效质量扩散系数、对流质量传递系数、柯西边界条件、有限圆柱体

引言

长叶姜是一种原产于印度的植物, 既被用作香料, 也用于提取其提取物。由于其成分具有抗氧化、抗炎、抗突变、抗癌和抗菌等功效, 因此关于这种农作物的研究在文献中非常丰富 (Negi 等人, 1999 年; Kim 等人, 2000 年; Quiles 等人, 2002 年; Surh, 2002 年; Inano 和 Onoda, 2002 年; Mohanty 等人, 2004 年; Prasad 等人, 2006 年; Xia 等人, 2007 年; Adaramoyea 等人, 2009 年; Chen 等人, 2010 年; Yue 等人, 2010 年)。一般来说, 当这种植物被收获时, 其根茎中的水分含量很高。根据 Silva 等人 (2007 年) 的研究, 水是新鲜食品中的一种重要成分, 其浓度很高, 对食品的口感、消化过程以及物理结构都有显著影响。实际上, 食品中的几乎所有变质过程都受到其中水分浓度和流动性等因素的影响。因此, 在收获之后、在储存或运输之前, 通常都会对根茎进行干燥处理。有一些研究探讨了姜黄的干燥过程。例如, Prasad 在 2009 年的研究中描述了在阳光直射下姜黄的干燥过程, 并研究了该过程的热力学特性。在这项研究中, 研究人员探讨了产品表面温度与周围环境温度之间的差异对对流热传递系数的影响。另一方面, Prathapan 等人也在 2009 年研究了热处理对新鲜姜黄根茎中的姜黄素、颜色值以及总多酚含量的影响。根据 Devahastin 和 Niamnuy 在 2010 年的研究, 在干燥过程中不仅会发生营养方面的变化, 还会发生其他方面的变化, 包括物理性质和微观结构的变化。这些变化非常重要, 需要加以优化, 最好通过采用各种建模方法来实现这一目标。

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approaches. Naturally, the own drying process needs be optimized and, for this, a model can be useful.

In order to describe a drying process, mathematical models are frequently used. Among these models, the liquid diffusion is very found in the drying description of agricultural products (Mulet et al. 2005; Hacıhafızoglu et al. 2008; Silva et al. 2009, 2010a). In order to solve the diffusion equation, in many of these researches the boundary condition of the first kind is assumed in the solution. In addition, the effective mass diffusivity and the volume of the solid are considered constant during the whole process (Mulet et al. 2005; Hacıhafızoglu et al. 2008; Silva et al. 2009). For these assumptions, considering a simple geometry, the diffusion equation can be solved analytically (Mulet et al. 2005; Hacıhafızoglu et al. 2008; Silva et al. 2009, 2012). On the other hand, if the boundary condition is of the third kind, normally the diffusion equation is solved numerically (Wu et al. 2004; Mulet et al. 2005; Olek and Weres 2007; Da Silva et al. 2012), although a few analytical solutions are also found in the literature (Silva et al. 2010a, b). It is worth pointing out that in some works, as in Mulet et al. (2005), two analytical solutions are presented for the boundary condition of the first kind, and one numerical solution is proposed for the boundary condition of the third kind. According Silva et al. (2010b), a possible reason for the lack of analytical solutions in the literature for diffusion problems with boundary condition of the third kind could be the large quantity of roots of the characteristic equation that must be determined in order to cover the entire domain of the mass transfer Biot number, when many terms of the series are taken account.

For the *Curcuma longa* rhizomes, the drying time depends on the size of the cut pieces and also if the product was peeled (removed peridermis) or not. For instance, according Mulet et al. (2005), peeled rhizomes, cylindrical-shaped, with 10.0 mm of height and radius of 5.0 mm are dried from dimensionless moisture content of 1 to 0.2 into 9,400 s. If the pieces contain shell (with peridermis, and therefore the radius is 6.5 mm), the drying time is 23,900 s. Thus, it seems to be economically advantageous to dry the peeled product in small pieces. In their researches, Mulet et al. (2005) assumed boundary condition of the first kind in order to describe the drying kinetics of peeled rhizomes. However despite the good statistical indicators obtained in the simulation of drying kinetics, a visual inspection of the curve seems to indicate a biased fit. Thus, it appears appropriate to examine whether the boundary condition of first kind for the diffusion equation is really adequate to describe the drying process.

The main objective of this article is to analyze what is the appropriate boundary condition for the diffusion equation used to describe drying of peeled rhizomes of *Curcuma longa*, cut into a cylindrical shape. With this objective, analytical solutions are coupled to an optimizer in order to

identify the better boundary condition. Once the appropriate boundary condition is identified and the process parameters are determined for an experimental dataset, the whole drying is described in detail.

Material and methods

In order to obtain the analytical solution for the diffusion equation in a finite cylinder, the following assumptions should be established: the cylinder must be considered homogeneous and isotropic; the moisture distribution within the cylinder should be initially uniform, with axial symmetry; the properties of the hot air used in drying should remain constant during the whole process; the only drying mechanism is liquid diffusion; the dimensions of the finite cylinder do not vary during the process; the effective mass diffusivity and the convective mass transfer coefficient do not vary during the process.

Analytical solution

A finite cylinder of radius R and height L is shown in Fig. 1. A position within the cylinder can be given by the coordinates (r, y) defined through the system of axes r and y with origin at the centre of the cylinder.

In order to describe a drying process, the diffusion equation can be written as follows:

$$\frac{\partial X}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(rD \frac{\partial X}{\partial r} \right) + \frac{\partial}{\partial y} \left(D \frac{\partial X}{\partial y} \right), \quad (1)$$

where X is the moisture content (dry basis), D is the effective mass diffusivity, t is the time, r and y define a position within cylinder. On the other hand, the boundary condition of the third kind, also called Cauchy boundary condition, is defined by:

$$-D \frac{\partial X(r, y, t)}{\partial r} \Big|_{r=R} = h[X(r, y, t) \Big|_{r=R} - X_{eq}], \quad (2)$$

and

$$-D \frac{\partial X(r, y, t)}{\partial y} \Big|_{y=\pm L/2} = h[X(r, y, t) \Big|_{y=\pm L/2} - X_{eq}], \quad (3)$$

where h is the convective mass transfer coefficient and X_{eq} is the equilibrium moisture content (dry basis). Equation (2) refers to an infinite cylinder with radius R while Eq. (3) refers to an infinite slab with thickness L . The composition of these two simple geometries generates the finite cylinder. In these two equations it was imposed the same value h , which means same resistance to the water flux in every surfaces.

方法。当然, 自身的干燥过程需要得到优化, 而为此, 使用一个模型会非常有帮助。

为了描述干燥过程, 通常会使用数学模型。在这些模型中, 液体扩散现象在农产品干燥过程中被广泛考虑 (Mulet 等人, 2005 年; Hacıhafızoglu 等人, 2008 年; Silva 等人, 2009 年、2010 年)。为了解决扩散方程, 许多研究在求解过程中都假设了第一类边界条件。此外, 人们认为在整个过程中, 固体的有效质量扩散系数和体积是恒定的 (Mulet 等人, 2005 年; Hacıhafızoglu 等人, 2008 年; Silva 等人, 2009 年)。基于这些假设, 在简单几何形状的情况下, 扩散方程可以解析求解 (Mulet 等人, 2005 年; Hacıhafızoglu 等人, 2008 年; Silva 等人, 2009 年、2012 年)。另一方面, 如果边界条件是第三类, 那么通常需要数值方法来解决扩散方程 (Wu 等人, 2004 年; Mulet 等人)。在 2005 年, Olek 和 Weres 在 2007 年, 以及 Da Silva 等人于 2012 年发表了相关研究成果。不过, 在文献中也有些解析解的方法 (Silva 等人, 2010a, b)。值得注意的是, 在一些研究中, 如 Mulet 等人 (2005 年) 的研究中, 对于第一类边界条件, 提出了两种解析解; 而对于第三类边界条件, 则只提出了一种数值解。根据 Silva 等人 (2010b) 的观点, 文献中缺乏第三类边界条件下的解析解的原因可能是: 为了涵盖整个质量传递 Biot 数域, 需要确定特征方程的大量根, 而这通常需要考虑大量的项。

对于长叶根茎来说, 干燥时间取决于切割后的块大小, 以及产品是否已经去皮 (即去除了表皮层)。例如, 根据 Mulet 等人 (2005 年的研究), 如果根茎是圆柱形的, 高度为 10.0 毫米, 半径为 5.0 毫米, 并且水分含量从 1% 降至 0.2%, 那么干燥时间约为 9,400 秒。如果根茎带有外壳 (即仍然有表皮层, 因此半径变为 6.5 毫米), 那么干燥时间则变为 23,900 秒。因此, 将去皮后的根茎切成小块进行干燥似乎在经济上是更合理的选择。在研究中, Mulet 等人假设了第一类边界条件来描述去皮根茎的干燥动力学过程。不过, 尽管在模拟过程中获得了良好的统计指标, 但从曲线来看, 这种拟合方式似乎存在偏差。因此, 有必要探讨扩散方程的初值条件是否真的足以描述干燥过程。

这篇文章的主要目标是分析用于描述长叶三七的剥制根茎干燥过程的扩散方程的合适边界条件。具体来讲, 就是确定适用于该扩散方程的条件。

分析解需要与优化算法相结合, 以便确定最佳的边界条件。一旦确定了合适的边界条件, 并且为实验数据集确定了相关工艺参数后, 整个干燥过程就能得到详细的描述。

材料与方法

为了获得有限圆柱体中的扩散方程的解析解, 需要做出以下假设: 圆柱体被认为是均匀且各向同性的; 圆柱体内的水分分布在初始时应该是均匀的, 并且具有轴对称性; 用于干燥的热空气的性质在整个过程中应保持恒定; 唯一的干燥机制是液体扩散; 在过程中, 有限圆柱体的尺寸不会发生变化; 在过程中, 质量扩散的有效系数和对质量传递的传递系数也不会发生变化。

解析解

如图 1 所示, 这是一个半径为 R 、高度为 L 的有限圆柱体。圆柱体内部的一个点可以通过以圆柱体中心为原点, 用 r 和 y 轴来表示的坐标 (r, y) 来指定。

为了描述干燥过程, 扩散方程可以表述为如下形式:

$$\frac{\partial X}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r D \frac{\partial X}{\partial r} \right) + \frac{\partial}{\partial y} \left(D \frac{\partial X}{\partial y} \right); \quad (1)$$

其中, X 表示水分含量 (干基值), D 表示有效质量扩散系数, t 表示时间; 而 r 和 y 则用于定义圆柱体内的某个位置。另一方面, 第三种边界条件也被称为柯西边界条件, 其定义方式如下:

$$D \frac{\partial X}{\partial r} = h (X - X_{\infty}) \quad \text{在 } r=R \text{ 处}; \quad (2)$$

以及

$$D \frac{\partial X}{\partial y} = h (X - X_{\infty}) \quad \text{在 } y=L/2 \text{ 处}; \quad (3)$$

其中, h 表示对流传质系数, X_{∞} 则表示平衡水分含量 (干基)。方程 (2) 描述的是半径为 R 的无限长圆柱体, 而方程 (3) 则描述的是厚度为 L 的无限长平板。这两种简单几何形状的组合就构成了有限长圆柱体的模型。在这两个方程中, 都采用了相同的 h 值, 这意味着所有表面的水通量阻力都是相同的。

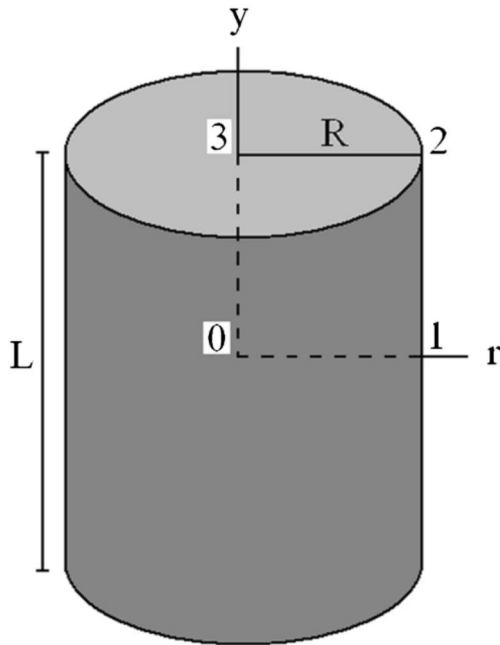


Fig. 1 Finite cylinder (radius R and height L)

The analytical solution of Eq. (1) in the position (r,y) at an instant t , with the boundary conditions given by Eqs. (2) and (3), is obtained by separation of variables (Luikov 1968; Crank 1992):

$$X(r,y,t) = X_{eq} - (X_{eq} - X_i) \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} A_{n,1} A_{m,2} J_0(\mu_{n,1} \frac{r}{R}) \cos(\mu_{m,2} \frac{y}{L/2}) \times \exp\left[-\left(\frac{\mu_{n,1}^2}{R^2} + \frac{\mu_{m,2}^2}{(L/2)^2}\right)Dt\right], \quad (4)$$

in which X_i is the initial moisture content. The coefficients $A_{n,1}$ e $A_{m,2}$ are defined as follows:

$$A_{n,1} = \frac{2Bi_1}{J_0(\mu_{n,1})(Bi_1^2 + \mu_{n,1}^2)}, \quad (5)$$

and

$$A_{m,2} = (-1)^{m+1} \frac{2Bi_2(Bi_2^2 + \mu_{m,2}^2)^{1/2}}{\mu_{m,2}(Bi_2^2 + Bi_2 + \mu_{m,2}^2)}, \quad (6)$$

in which Bi_1 and Bi_2 are the mass transfer Biot numbers for the infinite cylinder and infinite slab, respectively, and are given by

$$Bi_1 = \frac{hR}{D}, \quad (7)$$

and

$$Bi_2 = \frac{h(L/2)}{D}. \quad (8a)$$

In Eq. (5), J_0 is the Bessel function of first kind and zero order.

An expression for h can be obtained from Eq. (7), and this expression can be substituted into Eq. (8a). Thus, the mass transfer Biot number for the infinite slab can be given in the following way:

$$Bi_2 = \frac{Bi_1(L/2)}{R}. \quad (8b)$$

In Eq. (4), (5) and (6), $\mu_{n,1}$ and $\mu_{m,2}$ are respectively the roots of the characteristic equations for the infinite cylinder and infinite slab. The expressions for such equations are:

$$\frac{J_0(\mu_{n,1})}{J_1(\mu_{n,1})} = \frac{\mu_{n,1}}{Bi_1}, \quad (9)$$

and

$$\cot \mu_{m,2} = \frac{\mu_{m,2}}{Bi_2}. \quad (10)$$

In Eq. (9), J_1 is the Bessel function of first kind and first order.

The expression to determine the average moisture content at time t can be given in the following way:

$$\bar{X}(t) = \frac{1}{V} \int X(r,y,t) dV. \quad (11)$$

where V is the volume of the finite cylinder.

Substituting Eq. (4) into Eq. (11), the following result is obtained:

$$\bar{X}(t) = X_{eq} - (X_{eq} - X_i) \times \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} B_{n,1} B_{m,2} \exp\left[-\left(\frac{\mu_{n,1}^2}{R^2} + \frac{\mu_{m,2}^2}{(L/2)^2}\right)Dt\right], \quad (12)$$

in which $\bar{X}(t)$ is the average moisture content at time t , in dry basis.

The coefficient $B_{n,1}$ is given by:

$$B_{n,1} = \frac{4Bi_1^2}{\mu_{n,1}^2(Bi_1^2 + \mu_{n,1}^2)}, \quad (13)$$

and the coefficient $B_{m,2}$ is defined by the expression:

$$B_{m,2} = \frac{2Bi_2^2}{\mu_{m,2}^2(Bi_2^2 + Bi_2 + \mu_{m,2}^2)}. \quad (14)$$

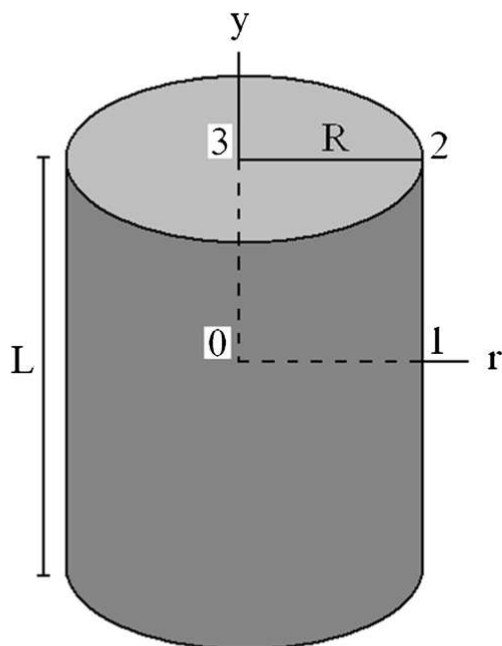


图 1 有限圆柱体 (半径为 R, 高度为 L)

方程(1)在时刻 t 时的位置 (r, y) 处的解析解, 通过变量分离法可以得到 (参见 Luikov 1968 年; Crank 1992 年的研究成果)。

$$X(r, y, t) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} A_{nm} J_0(\mu_n r) \exp\left(-\frac{\mu_n^2}{R^2} t\right) \cos\left(\frac{\mu_m y}{L}\right) \quad (8a)$$

其中 X 表示初始水分含量。系数 A 和 A_e 的定义如下:

$$A_{nm} = \frac{2Bi_n}{J_0(\mu_n)} \exp\left(-\frac{\mu_n^2}{R^2} t\right) \quad (8b)$$

以及

$$A_{nm} = \frac{2Bi_n}{\mu_n^2} \exp\left(-\frac{\mu_n^2}{R^2} t\right) \quad (8c)$$

其中, Bi_n 和 Bi_m 分别是无限圆柱体和无限薄板的质量传递 Biot 数, 其表达式如下:

$$Bi_n = \frac{hR}{D} \quad (8d)$$

以及

$$Bi_m = \frac{hL}{D} \quad (8e)$$

在等式(5)中, J 是第一类、零阶的贝塞尔函数。

可以从方程(7)中得到 h 的表达式, 然后将该表达式代入方程(8a)中。因此, 无限厚平板的质量传递 Biot 数可以如下计算:

$$Bi = \frac{hL}{D} \quad (8f)$$

在方程(4)、(5)和(6)中, μ_n 和 μ_m 分别是无限圆柱体和无限薄板对应的特征方程的根。这些方程的表达式如下:

$$J_0(\mu_n) = 0 \quad (8g)$$

以及

$$\cot \mu_m = \frac{\mu_m}{Bi} \quad (8h)$$

在等式(9)中, J 是第一类一阶的贝塞尔函数。

用于确定时间 t 时的平均水分含量的表达式可以如下所示:

$$\bar{X}(t) = \frac{1}{V} \int_V X(r, y, t) dV \quad (8i)$$

其中, V 表示有限圆柱体的体积。

将方程(4)代入方程(11)后, 可以得到以下结果:

$$\bar{X}(t) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} A_{nm} \exp\left(-\frac{\mu_n^2}{R^2} t\right) \cos\left(\frac{\mu_m y}{L}\right) \quad (8j)$$

其中, $\bar{X}(t)$ 表示时间 t 时的平均水分含量, 单位为干基单位。

系数 B 的表达式为:

$$B_{nm} = \frac{4Bi_n}{\mu_n^2} \exp\left(-\frac{\mu_n^2}{R^2} t\right) \quad (8k)$$

以及由以下表达式定义的系数 Bis :

$$B_{nm} = \frac{4Bi_n}{\mu_n^2} \exp\left(-\frac{\mu_n^2}{R^2} t\right) \quad (8l)$$

An auxiliary program was developed in Fortran, and the first 16 roots of Eq. (9) were calculated for 452 specified values of mass transfer Biot numbers from $Bi_1=0$ (which corresponds to an infinite resistance of water flux at the surface) up to $Bi_1=200$ (which practically corresponds to an equilibrium boundary condition). Similarly, in another program, the first 16 roots of Eq. (10) were calculated for 472 specified values of mass transfer Biot numbers from $Bi_2=0$ up to $Bi_2=200$. Note that a dataset with 14,784 roots (7,232 for the infinite cylinder and 7,552 for the infinite slab) was created in order to extract results from Eqs. (4) and (12).

For an infinite cylinder, the expression for the mass transfer Biot number was defined by Eq. (7). On the other hand, for a specified value of Bi_1 , the value of Bi_2 can be calculated from Eq. (8b). If the roots corresponding to Bi_2 are not available into the dataset previously defined with 7,552 values for the infinite slab, they can be calculated by linear interpolation, using the available values. Then, for a given mass transfer Biot number Bi_1 (infinite cylinder), the correspondent Bi_2 (infinite slab) is also defined and, consequently, the roots $\mu_{n,1}$ and $\mu_{m,2}$ are determined as was discussed. So, the coefficients $A_{n,1}$, $A_{m,2}$, $B_{n,1}$, and $B_{m,2}$ are calculated using Eqs. (5), (6), (13) and (14), respectively. Thus, Eqs. (12) and (4) can be used to determine, respectively, $\bar{X}(t)$ and $X(r, y, t)$ or the correspondent dimensionless values.

Equation (12) can be rewritten to express the dimensionless moisture content, which is defined as follows:

$$\bar{X}^*(t) = \frac{\bar{X}(t) - X_{eq}}{X_i - X_{eq}} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} B_{n,1} B_{m,2} \exp \left[- \left(\frac{\mu_{n,1}^2}{R^2} + \frac{\mu_{m,2}^2}{(L/2)^2} \right) Dt \right]. \quad (15)$$

In this paper, the experimental data obtained for drying kinetics will be analyzed in the dimensionless form, as defined in Eq. (15).

Determination of the process parameters

In order to determine D and h for an experimental dataset, the algorithm developed by Silva et al. (2010a) will be adapted for a finite cylinder. The objective function to be minimized is given by (Bevington and Robinson 1992; Taylor 1997):

$$\chi^2 = \sum_{i=1}^{N_p} [\bar{X}_i^{\text{exp}} - \bar{X}_i^{\text{ana}}(D, Bi_1)]^2 \frac{1}{\sigma_i^2} \quad (16)$$

where $\bar{X}_i^{\text{exp}}$ is the i^{th} experimental point of the average moisture content; $\bar{X}_i^{\text{ana}}(D, Bi_1)$ is the average moisture content at the same point, calculated from Eq. (12) with

256 terms (16×16); σ_i is the standard deviation of the experimental average moisture content at the point i ; D is the effective mass diffusivity; and N_p is the number of experimental points. For a specified value of Bi_1 (and consequently Bi_2), the chi-square depends only on a single variable, namely the effective mass diffusivity. Thus, optimum values of D can be determined for each one of the 452 Bi_1 defined for the cylinder. The better D is determined by the minimum χ^2 among the 452 minima calculated. Each one of the 452 optimization process was accomplished as recommended by Silva et al. (2009). According these authors, an initial value close to zero (1×10^{-20}) is attributed to D , and replaced into Eq. (12). Thus, $\bar{X}_i^{\text{ana}}$ can be calculated for a given time and, consequently, χ^2 can be determined for a set of experimental data through Eq. (16). Then, the value of D is doubled, and a new χ^2 is calculated. The new χ^2 is compared with the former value. If the new value is lower than the foregoing, D is again doubled, the corresponding value of χ^2 is calculated, and compared with the former one. This procedure is repeated until the last calculated χ^2 is greater than the anterior value. Note that, even starting with a modest initial value for the D , the subsequent values grow geometrically, ensuring a quick optimization process. On the other hand, the antepenultimate and ultimate values of D , denoted as D_a and D_b , respectively, define a region which contains the minimal value of χ^2 . The penultimate value of the effective mass diffusivity corresponds to the smallest value of χ^2 obtained in this interval. The latter procedure can be refined between D_a and D_b , subdividing this interval in n_v values of D uniformly distributed. Then, a more refined interval can be obtained, and this procedure can be repeated until a convergence criterion is satisfied. On the other hand, if the statistical weights $1/\sigma_i^2$ were not obtained from the experiment and are, therefore, unknown, a constant value (for instance, $1/\sigma_i^2 = 1$) should be artificially attributed to all experimental points, meaning the same statistical weight for all of them.

The described optimizer was coupled to Eq. (12), and the source code was developed in a computer with Intel Pentium IV and 2 GB RAM. The source code was compiled in Compaq Visual Fortran (CVF) 6.6.0 Professional Edition, using the programming option QuickWin Application, and the platform Windows Vista. Criterion of convergence was stipulated in this article as 1×10^{-16} . Note that if the dataset are given in the dimensionless form, Eq. (15) must be used instead Eq. (12).

Once D and h are determined by optimization, Eq. (4) is used to calculate the moisture distribution as function of the position (r, y), at time t . Graphs representing the drying kinetics and contour plots showing the moisture distributions in specified times are also carried out by the developed software.

用 Fortran 语言编写了一个辅助程序, 该程序计算了方程(9)的前 16 个根。这些根是针对 452 个不同的 Biot 数值计算得出的, 这些数值从 Bi_{00} (相当于表面水流的无限阻力) 到 Bi_{0200} (实际上相当于平衡边界条件)。同样, 在另一个程序中, 方程(10)的前 16 个根也被计算出来, 这些根是针对 472 个不同的 Biot 数值得出的, 数值范围从 Bi_{00} 到 Bi_{0200} 。需要注意的是, 为了从方程(4)和(12)中得到结果, 还生成了一个包含 14,784 个根的数据集 (其中 7,232 个用于无限圆柱情况, 7,552 个用于无限平板情况)。

对于无限大的圆柱体, 质量传递 Biot 数的表达式由方程(7)给出。另一方面, 对于特定的 Biot 数值, 可以通过方程(8b)来计算该数值。如果数据集中没有对应于 Biot 数的根值, 那么可以使用线性插值法, 利用已有的数值来计算这些根值。然后, 对于给定的质量传递 Biot 数 Bi (无限大圆柱体), 相应的 Biot 数 Bi (无限大平板) 也可以被确定下来, 从而可以像之前讨论的那样确定出 μ 和 μ 的值。最后, 系数 A 、 A 、 B 和 B 可以通过方程(5)、(6)、(13)和(14)来计算。因此, 方程(12)和(4)可以用来分别确定 $X_{\delta tP}$ 和 $X_{\delta r; y; tP}$, 或者相应的无量纲值。

方程(12)可以重新表述为表示无量纲水分含量, 其定义如下:

$$\frac{X_{\delta tP} - X}{X_{\text{投影仪}} - X} = \frac{X_{11} - X}{X_{m/41} - X} \quad \text{BB 经验 概率} \quad \frac{\mu}{R} \quad \mu \quad \frac{\mu}{\partial L} \quad 2 \Rightarrow \quad \delta 15p$$

在本文中, 关于干燥动力学的实验数据将被以无量纲形式进行分析, 这种无量纲形式是根据方程(15)定义的。

确定工艺参数

为了确定实验数据集中的 D 和 h 值, 需要采用 Silva 等人 (2010a) 开发的算法来处理有限尺寸的圆柱体问题。需要最小化的目标函数可以表示为 (Bevington 和 Robinson 1992; Taylor 1997):

$$c^{1/4} \quad X^N \quad X \quad \text{投影到 } X_{i\delta t}; \text{ 生物 } \frac{1}{\sigma} \quad \delta 16p$$

其中, X_i 表示平均水分含量中的第 i 个实验值; X_D , X_{Bi} 这里的 “ p ” 指的是同一位置的平均水分含量, 根据方程(12)进行计算得出。

共有 256 个参数 (16×16 的矩阵形式); σ 表示在给定时间点实验测得的平均水分含量的标准差; D 为有效质量扩散系数; N 为实验点的数量。对于特定的 Bi 值, 卡方分布仅依赖于一个变量, 即有效质量扩散系数。因此, 可以通过计算 452 个不同 Bi 值对应的 D 值, 来确定每个 Bi 值下的最优 D 值。最优 D 值由计算出的 452 个最小值中最小的那个决定。整个优化过程都是按照 Silva 等人 (2009 年) 的建议进行的。根据他们的方法, 初始值 D 被设定为接近零的值 (1×10), 然后将其代入方程 (12) 中。这样, 就可以为给定的时间计算出 X 值, 进而通过方程 (16) 确定一组实验数据的 χ 值。之后, 将 D 值翻倍, 重新计算 χ 值, 并将新得到的 χ 值与之前的值进行比较。如果新的值低于之前的值, 那么 D 会再次翻倍, 然后计算出相应的 χ 值, 并将其与之前的值进行比较。这个过程会一直重复, 直到最后算得的 χ 值大于之前的值为止。需要注意的是, 即使从较低的 D 值开始, 后续的值也会以几何级数增长, 从而确保优化过程能够快速进行。另一方面, D 的倒数第二和最后的值分别记为 D 和 D , 它们所围成的区间包含了 χ 的最小值。有效质量扩散率的倒数第二的值, 就是在这个区间内获得的 χ 的最小值。这个步骤可以在 D 和 D 之间进一步细分, 将区间划分为 n 个等分布的 D 值。然后, 可以得到更精细的区间, 并重复这个过程, 直到满足收敛条件为止。另外, 如果统计权重为 1σ

这些数值并非通过实验获得的, 因此属于未知值; 它们是一个常数值 (例如, 1σ)。

1/4 1) 应该将所有实验点的数据都进行人为分配, 也就是说, 所有实验点具有相同的统计权重。

所描述的优化算法与方程(12)相结合使用。源代码是在配备有英特尔奔腾 IV 处理器和 2GB 内存的计算机上编写的。源代码使用 Compaq Visual Fortran (CVF) 6.6.0 专业版编译器进行编译, 编程工具为 QuickWin Application, 运行环境为 Windows Vista 操作系统。本文中, 收敛标准被设定为 1×10 。需要注意的是, 如果数据集以无量纲形式给出, 那么应该使用方程(15)而不是方程(12)来进行计算。

一旦通过优化方法确定了 D 和 h 的值后, 就可以使用方程(4)来计算在时间 t 时, 各位置 (r, y) 处的水分分布。该软件还能绘制出干燥过程的动力学曲线, 以及在特定时间点的水分分布图。

Material

An experimental data set obtained by Mulet et al. (2005) referring to the drying kinetics of *Curcuma longa* rhizomes will be analyzed in this paper, in order to test the two diffusion models to describe the process. In one of their experiments, Mulet et al. (2005) used peeled rhizomes, cut in cylindrical pieces with height $L=10.0$ mm and radius $R=5.0$ mm. The temperature of the hot air was kept in 70°C and its velocity was 4.5 ms^{-1} . The obtained data set were analyzed by the authors in the dimensionless form, using Eq. (15). Because the pieces are peeled, the authors considered that the boundary condition of the first kind, also named Dirichlet boundary condition, was adequate to describe the drying process.

In the experiment, Mulet et al. (2005) did not determine the statistical weights of the experimental points (dimensionless average moisture contents) and, in the present article, these statistical weights will be made equal to 1. In order to determine the equilibrium moisture content, the relationship between water activity and moisture content of *Curcuma longa* rhizomes was obtained using a NOVASINA TH/RTD 200 electric hygrometer (Switzerland). The Henderson model was considered to describe the desorption equilibrium isotherm.

For the drying conditions employed in the experiment, liquid diffusion models appear to be reasonable, despite a possible anisotropy and heterogeneity of the sample. In this paper, it will be supposed that these characteristics occur in a little scale and can, therefore, be discarded as well as the shrinkage (Baronas et al. 2001; Kulasiri and Woohead 2005; Mulet et al. 2005; Ricardez et al. 2005; Olek and Weres 2007; Silva et al. 2010b).

Results and discussion

Equation (15) was fitted to the experimental dataset and the obtained results, considering boundary condition of the first (model 1) and third (model 2) kinds, are presented in the

following. Comparison between the two models is made using the statistical indicators chi-square (χ^2) and determination coefficient (R^2); and also average error and error distribution about the simulated curves.

Model 1: boundary condition of the first kind

The optimization process considering no resistance at the surfaces (mass transfer Biot number with a known value, stipulated as $Bi_1=200$) and effective diffusivity unknown (boundary condition of first kind) results in: $D=3.46\times 10^{-10}\text{ m}^2\text{ s}^{-1}$; $R^2=0.9988$ and $\chi^2=1.7095\times 10^{-3}$. The relation between chi-square and the effective mass diffusivity can be observed in Fig. 2a, in the vicinities of the optimal point. Using the value obtained for the effective mass diffusivity, the drying kinetics simulation together with the experimental data can be shown through Fig. 2b.

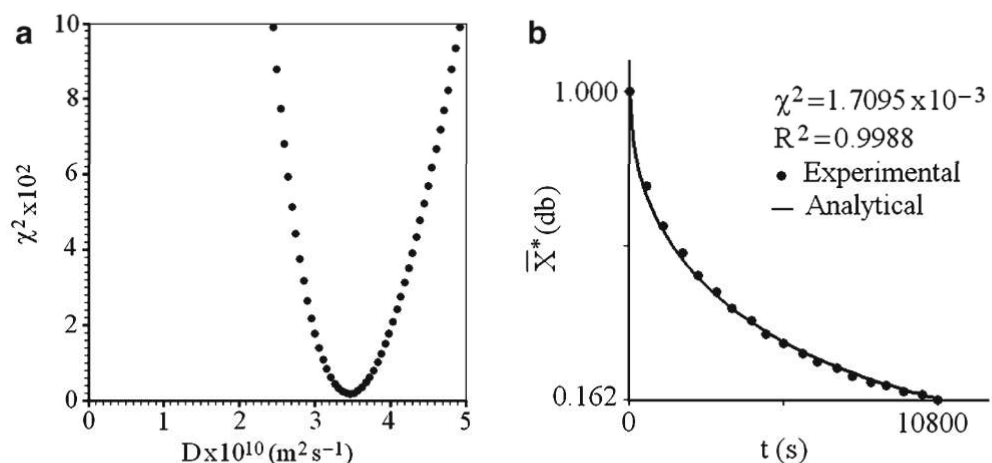
The obtained statistical indicators appear to be reasonable. However, Fig. 2b indicates that the first experimental points are above the simulated curve while the last ones are below this curve. This means that there is not a random distribution of the experimental points about the simulated curve. So, defining the error of each point i by

$$\text{error}_i = \bar{X}_i^{\text{exp}} - \bar{X}_i^{\text{ana}}, \quad (17)$$

the dispersion graphs for model 1 can be visualized in Fig. 3.

An inspection of Fig. 3 enables to affirm that the errors are correlated with the values of the experimental average moisture content. For instance, it is possible fit a polynomial of the third degree to the points $(\bar{X}^{\text{exp}}, \text{error})$ with certain success. So, despite good statistical indicators resulting of model 1, the adequacy of a second model will be investigated in this article.

Fig. 2 **a** Chi-square in the vicinities of the optimal point; **b** Drying kinetics of *Curcuma longa* rhizomes for the boundary condition of the first kind



材料

本文将分析 Mulet 等人 (2005 年) 获得的关于长叶姜根茎干燥动力学的实验数据。他们使用两种扩散模型来描述这一过程。在他们的实验中, Mulet 等人使用了剥去外皮的根茎, 这些根茎被切割成高度为 $L010.0$ 毫米、半径为 $R05.0$ 毫米的圆柱形片段。热空气的温度被设定为 70°C , 其流速为 4.5 米每秒。作者使用方程 (15) 对获得的数据进行了无量纲化处理。由于根茎已经剥去外皮, 作者认为适用第一类边界条件, 即狄利克雷边界条件, 来描述干燥过程。

在实验中, Mulet 等人 (2005 年) 并未确定实验数据点的统计权重 (无量纲的平均水分含量), 在本文中, 这些统计权重将被设定为 1。为了确定平衡水分含量, 使用了 NOVASINATH/RTD 200 电子湿度计来测量 *Curcuma longa* 根茎的水活度与水分含量之间的关系。采用 Henderson 模型来描述脱附平衡等温线。

在实验中采用的干燥条件下, 尽管样品可能存在各向异性和不均匀性, 但液体扩散模型仍然被认为是合理的。在本文中, 我们假设这些特性发生在较小的尺度上, 因此可以忽略不计。此外, 关于收缩现象的问题 (Baronas 等人, 2001 年; Kulasiri 和 Woohead, 2005 年; Mulet 等人, 2005 年; Ricardez 等人, 2005 年; Olek 和 Weres, 2007 年; Silva 等人, 2010 年 b 卷), 也可以不予考虑。

以下是对这两种模型进行比较的结果。比较过程中使用了统计指标卡方检验和决定系数; 同时, 还考虑了模拟曲线的平均误差以及误差分布情况。

模型 1: 第一类边界条件

在忽略表面阻力 (质量传递的比奥数已知, 取值为 $Bi\ 0200$) 且有效扩散系数未知 (第一类边界条件) 的情况下进行优化计算, 得到如下结果: 扩散系数为 $D03.46 \times 10^{-10}$ 米每秒; 阻力值为 $R\ 00.9988$, 有效质量扩散系数为 $\chi01.7095 \times 10^{-3}$ 。在图 2a 中可以观察到卡方分布与有效质量扩散系数之间的关系, 在最优解附近。利用得到的有效质量扩散系数值, 可以通过图 2b 展示干燥动力学模拟结果以及实验数据。

所得到的统计指标看起来是合理的。然而, 如图 2b 所示, 前几个实验点的数值高于模拟曲线, 而最后几个点的数值则低于该曲线。这意味着实验点的分布并非随机的。因此, 可以认为每个点的误差为……

结果与讨论

方程(15)是根据实验数据集进行拟合得到的, 其中第一种模型 (模型 1)和第三种模型(模型 2)的边界条件也被考虑在了计算结果中。

错误: $\frac{1}{4} \times X \times i_j$

δ17p

模型 1 的扩散图可以在图 3 中看到。

通过对图 3 的检查, 可以确认这些误差与实验测得的平均水分含量值之间存在相关性。例如, 可以将第三阶多项式拟合到数据点 (X , 误差) 上, 且效果相当不错。因此, 尽管模型 1 的统计指标表现良好, 但本文还将对第二个模型的适用性进行进一步研究。

图 2 a 展示了在最优解附近的卡方分布情况; b 展示了长叶姜根茎的干燥动力学过程。

第一类边界条件

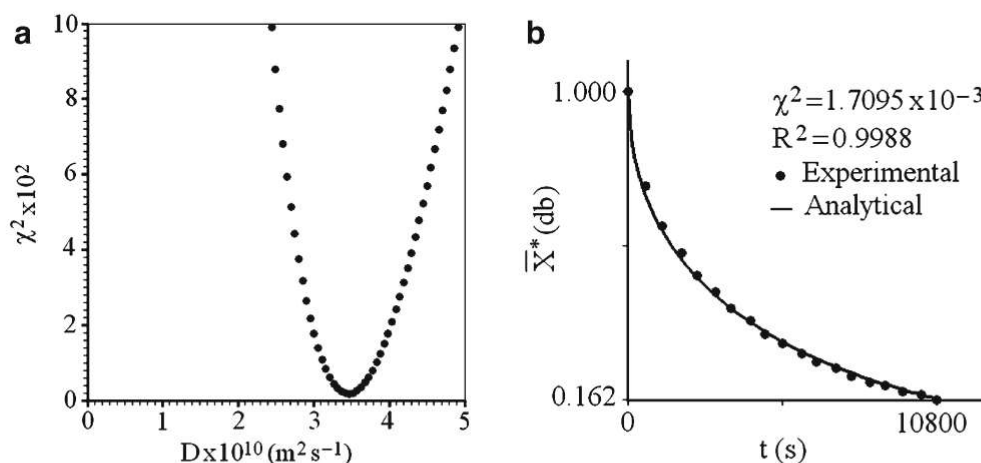
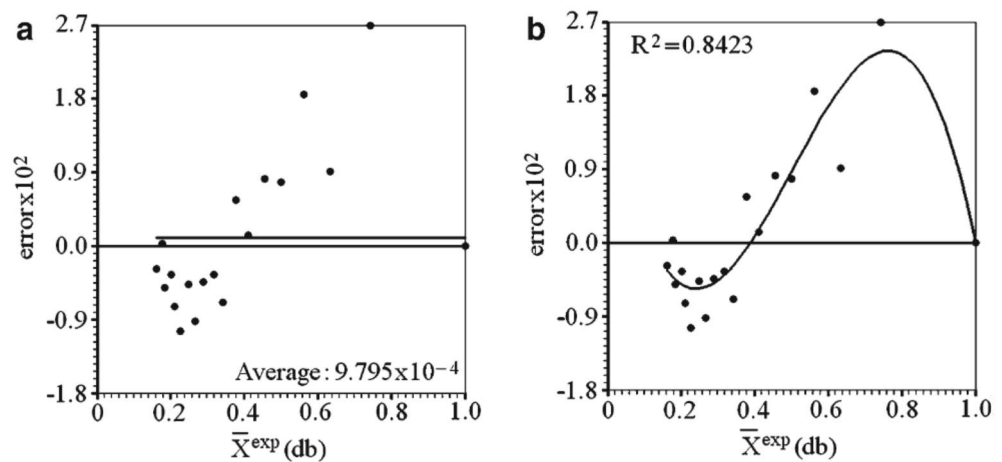


Fig. 3 Error dispersion for the drying kinetics with boundary condition of the first kind showing: **a** average error; **b** polynomial (third degree) fitting



Model 2: boundary condition of the third kind

An optimization process considering the mass transfer Biot number and the effective mass diffusivity as unknown values (boundary condition of the third kind) results in: $D = 3.85 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$; $h = 4.62 \times 10^{-6}$ ($Bi_1 = 60.00$); $R^2 = 0.9996$ and $\chi^2 = 3.5596 \times 10^{-4}$. The relation between chi-square and the mass transfer Biot number can be observed in Fig. 4a. Using the values obtained for D and h , the drying kinetics simulation together with the experimental data can be shown through Fig. 4b.

The dispersion graphs for model 2 can be visualized in Fig. 5.

The results obtained for the two optimization processes can be summarized in Table 1, which also presents the statistical indicators for the two models.

Using Eq. (4) and the parameters obtained for model 2, the drying kinetics referring to the points 1, 2 and 3 at the surface (Fig. 1) can be simulated and the results are shown through Fig. 6, which also shows the superposition of these three curves.

The moisture distributions in the rectangle 0123 (Fig. 1), at four specified instants, are shown in Fig. 7.

It is interesting to note that if Bi_1 and Bi_2 are considered infinite, Eq. (9) is written as $J_0(\mu_{n,1}) = 0$, and Eq. (10) as $\cot \mu_{m,2} = 0$ which results in $\mu_{m,2} = (2m - 1)\pi/2$, with $m = 1, 2, \dots, \infty$. Additionally, Eq. (13) results in $B_{n,1} = 4/\mu_{n,1}^2$, and Eq. (14) in $B_{m,2} = 2/\mu_{m,2}^2$. Thus, Eq. (12) is now the solution for the diffusion equation with boundary condition of the first kind (Luikov 1968).

For peeled rhizomes (in pieces), considering the boundary condition of the first kind, the results of the present article, using analytical solution, are identical to those obtained by Mulet et al. (2005). However, these authors did not investigate whether there might be some resistance to the water flux at the surfaces of cylindrical pieces. In fact, according to the present article, the consideration of the Cauchy boundary condition resulted in a mass transfer Biot number equal to 60. This value is really very large but suggests a certain resistance to the water flux at the boundaries, which could be considered.

For the Dirichlet boundary condition, Tab. 1 shows a chi-square about five times greater than the value obtained using the Cauchy boundary condition. In addition, for this last boundary condition, the error distribution about the

Fig. 4 **a** Chi-square in the vicinities of the optimal point; **b** Drying kinetics of *Curcuma longa* rhizomes for the boundary condition of the third kind

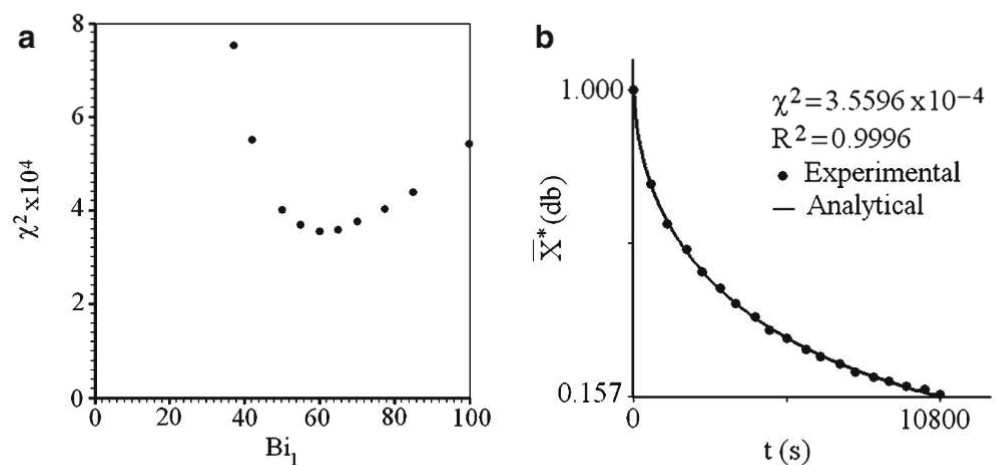
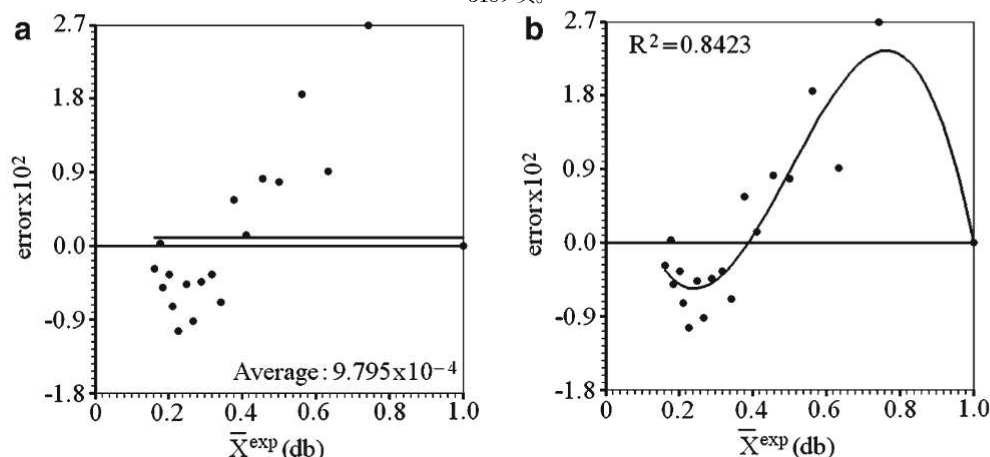


图 3 第一类边界条件下干燥动力学的误差分布情况: a 为平均误差; b 为多项式 (三阶) 拟合结果



模型 2: 第三种边界条件

在考虑质量传递 Biot 数和有效质量扩散系数为未知值 (边界条件为第三类) 的情况下, 优化过程得到的结果为: D_0 为 3.85×10^{-3} 米秒; h_0 为 4.62×10^{-6} ; R_{00} 为 0.9996; χ^2 为 3.5596×10^{-4} 。图 4a 展示了卡方分布与质量传递 Biot 数之间的关系。利用得到的 D 和 h 的值, 可以通过图 4b 来展示干燥动力学模拟结果以及实验数据。

模型 2 的扩散图可以在图 5 中看到。

两种优化过程所得到的结果可以在表 1 中汇总, 表 1 还列出了两种模型的统计指标。

利用公式(4)以及为模型 2 获得的参数, 可以模拟表面第 1、2、3 点的干燥动力学过程 (见图 1)。这些结果体现在图 6 中, 图 6 还展示了这三条曲线的叠加情况。

在矩形区域 0123 中, 四个特定时刻的湿度分布如图 7 所示。

有趣的是, 如果 B_i 和 B_i 被认为是无穷大的, 那么方程(9)可以表示为 $J(\mu)_{00}$, 而方程(10)则可以表示为 $\cot \mu_{00}$ 。这样就会得到 $\mu_0(2m-1)\pi/2$ 的结果, 其中 m 为 1, 2, …

…, ∞ 。此外, 方程(13)表明 $B \propto 4\mu$ 。以及

方程(14)在 $B \propto 2\mu$ 的情况下 因此, 方程(12)现在就是……

具有第一类边界条件的扩散方程的解 (Luikov 1968 年提出的方法)。

对于剥去外皮的根茎 (以块状形式提供), 考虑到第一类边界条件的限制, 本文使用解析方法得出的结果与 Mulet 等人 (2005 年) 的研究结果一致。不过, 这些研究者并未探讨圆柱形块状物表面是否可能存在对水流的阻力。实际上, 根据本文的研究, 考虑到柯西边界条件后, 质量传递 Biot 数的值达到了 60。这个数值相当大, 但表明在边界处确实存在对水流的阻力, 这一点值得考虑。

对于狄利克雷边界条件, 表 1 显示其误差大约是使用柯西边界条件所得到的值的五倍。此外, 对于这最后一个边界条件, 其误差分布情况如下:

图 4 a 展示了在最优值附近的卡方分布情况; b 展示了长叶姜根茎的干燥动力学过程。

第三类边界条件

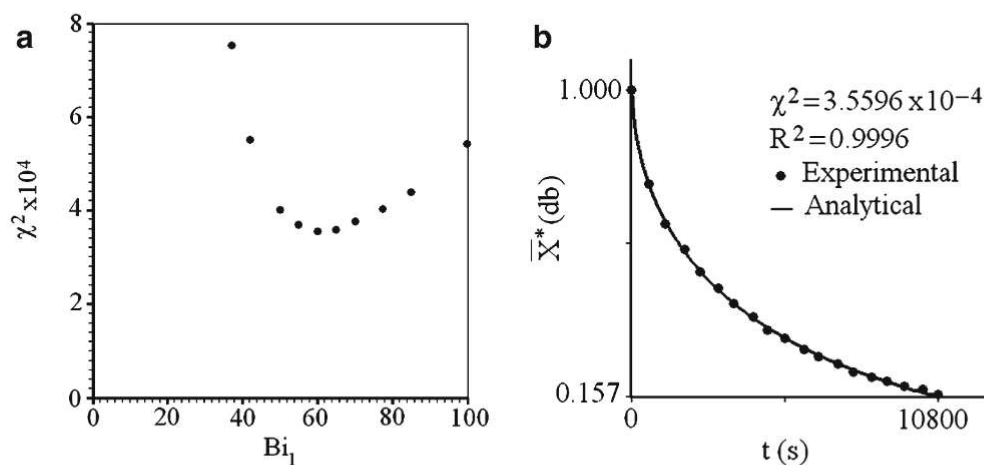
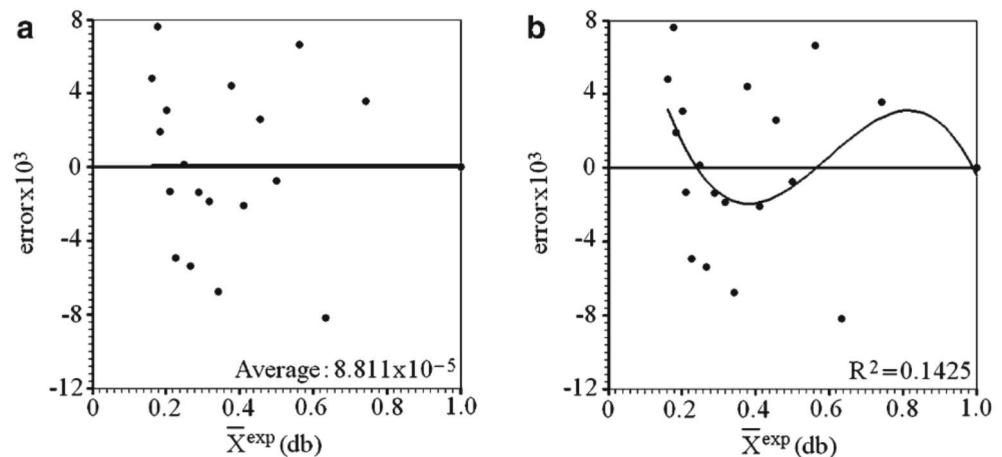


Fig. 5 Error dispersion for the drying kinetics with boundary condition of the third kind showing: **a** average error; **b** polynomial (third degree) fitting



simulated curve can be considered random (Fig. 5) while for the Dirichlet boundary condition the error distribution is biased (Fig. 3). A visual inspection in Figs. 3 and 5 makes it possible to affirm that model 2 results in a fit better than that model 1. Note that the determination coefficients of these curves presented in Figs. 3b and 5b reveal the same conclusion. In addition, for the Cauchy boundary condition, the average error is very close to zero, indicating that the model 2 is really superior to model 1.

Note that in different points at the surface highlighted in Fig. 1 (points 1, 2 and 3), the drying kinetics occurs in a different way (Figs. 6 and 7). Although the drying kinetics of the regions defined by the points 1 and 3 are virtually identical, they are significantly different from the drying kinetics for the region defined by the point 2 (Fig. 6). For the region defined by the point 2, almost instantaneously the local moisture content reaches the equilibrium value, but for the regions defined by the points 1 and 3, this affirmation is not true. This means that the small resistance to the water flux at the surface really is important to rigorously describe the drying process.

Due to some heterogeneity and anisotropy, the diffusion process in a given direction can be a little different from the diffusion in other direction. In addition, models analyzed do not consider the shrinkage and the variation of the diffusivity caused by the dimensional variation during the drying process. Thus, in this article, the values of the parameters determined by optimization and used in the simulation of the drying process really should be interpreted as effective values for the two investigated models.

It is worth note that the calculation of the roots of the characteristic equations for the finite cylinder (infinite

cylinder and infinite slab) and their organisation in tables accessible to the developed software is rather tedious, involving the determination of 14,784 values. However, the computational implantation of the algorithm is simple, which is a favourable characteristic of the procedure used in this article. Another aspect favourable is that the user only needs to inform the input set of experimental data because the optimization algorithm scans every the domain of the mass transfer Biot number and effective mass diffusivity.

As a final comment, the time for the execution of the developed software for finite cylinders strongly depends on the number of experimental data used for the realization of the optimization process. In the present study, for example, the total time for the determination of the parameters by fitting of the model 2 to 19 pairs $(t, \bar{X}^*)$ was 2.5 min.

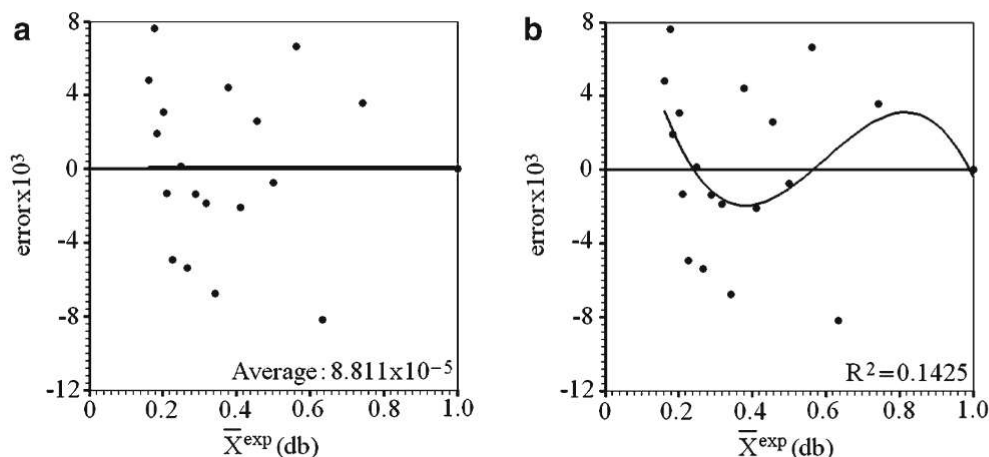
Conclusion

Model 1, commonly used in the literature to model the drying type approached in this article, reasonably describes the drying kinetics of peeled rhizomes (in pieces) of *Curcuma longa* (chi-square χ^2 equal to 1.7095×10^{-3} , and coefficient of correlation R^2 of 0.9988). However, the simulated curve is strongly biased. Better results are obtained with model 2, which considers the resistance to the water flux at the surface. In this case, despite the inclusion of a small resistance at the surface, the statistical indicators obtained for the drying kinetics are much better

Table 1 Process parameters and statistical indicators obtained for the two models

	D (m ² s ⁻¹)	h (ms ⁻¹)	$\chi^2 \times 10^3$	R ²	Average error $\times 10^4$	Error distribution
Model 1	3.46×10^{-10}	—	1.7095	0.9988	9.7950	Biased
Model 2	3.85×10^{-10}	4.62×10^{-6}	0.35596	0.9996	0.8811	Random

图 5 第三种边界条件下干燥动力学的误差分布情况: a 为平均误差; b 为三次多项式拟合结果



模拟曲线可以被视为随机的 (见图 5), 而狄利克雷边界条件下的误差分布则存在偏差 (见图 3)。通过图 3 和图 5 的直观观察可以确认, 模型 2 的拟合效果优于模型 1。注意, 图 3b 和图 5b 中所示这些曲线的决定系数也支持同样的结论。此外, 对于柯西边界条件而言, 平均误差几乎为零, 这表明模型 2 确实优于模型 1。

请注意, 在图 1 中标记的各个点 (点 1、点 2 和点 3) 处, 干燥过程的表现方式有所不同 (见图 6 和图 7)。虽然由点 1 和点 3 所确定的区域的干燥过程几乎是相同的, 但它们与由点 2 所确定的区域的干燥过程却有很大差异 (见图 6)。对于由点 2 所确定的区域, 局部水分含量几乎可以立即达到平衡值; 而由点 1 和点 3 所确定的区域则并非如此。这意味着, 表面对水分通量的较小阻力实际上对于准确描述干燥过程非常重要。

由于存在某些异质性和各向异性现象, 某一方向上的扩散过程可能会与其他方向上的扩散过程有所不同。此外, 所分析的模型并未考虑干燥过程中尺寸变化导致的收缩以及扩散性的变化。因此, 在本文中, 通过优化方法确定的参数值, 实际上应该被理解为两种被研究模型的有效参数值。

值得注意的是, 对于有限长度的圆柱体 (即无限长的情况), 其特征方程的根的计算方式有所不同。

在开发好的软件中, 对这些参数的组织和处理相当繁琐, 需要确定 14,784 个数值。不过, 该算法的计算实现非常简单, 这是本文所采用的该方法的一个优点。另一个优点是, 用户只需提供实验数据的输入集即可, 因为优化算法能够自动处理质量传递系数和有效质量扩散系数等参数。

最后要说的是, 针对有限圆柱体所开发的软件的执行时间, 很大程度上取决于用于优化过程的实验数据量。例如, 在本研究当中, 确定这些圆柱体的总时间就取决于所使用的实验数据量。



通过拟合模型 2 到 19 对 t ; X 来确定参数。

持续了 2.5 分钟。

结论

模型 1 在文献中常被用来模拟本文所描述的干燥过程。该模型能够合理地描述 *Curcuma longa* 的剥脱根茎的干燥动力学特性 (卡方检验的显著性为 1.7095×10^{-0} , 相关性系数为 0.9988)。不过, 模型 1 的模拟曲线存在较大偏差。而使用模型 2 则能得到更好的结果, 因为模型 2 考虑了表面对水分流动的阻力作用。在这种情况下, 尽管模型中包含了表面阻力这一小项, 但所得出的干燥动力学统计指标要优得多。

表 1 列出了两种模型所得到的工艺参数和统计指标。

D (毫秒)	h (毫秒)	$\chi \times 10$	R	平均误差 $\times 10$	错误分布
模型 1: 3.46×10^{-0} – 1.7095, 0.9988, 9.7950 有偏差的模型 2: 3.85×10^{-4} , 4.62×10^{-0} , 0.35596, 0.9996, 0.8811					
					随机的

Fig. 6 Local moisture content versus time at the region defined by: **a** point 1; **b** point 2; **c** point 3; **d** superposition of the curves (a), (b) and (c)

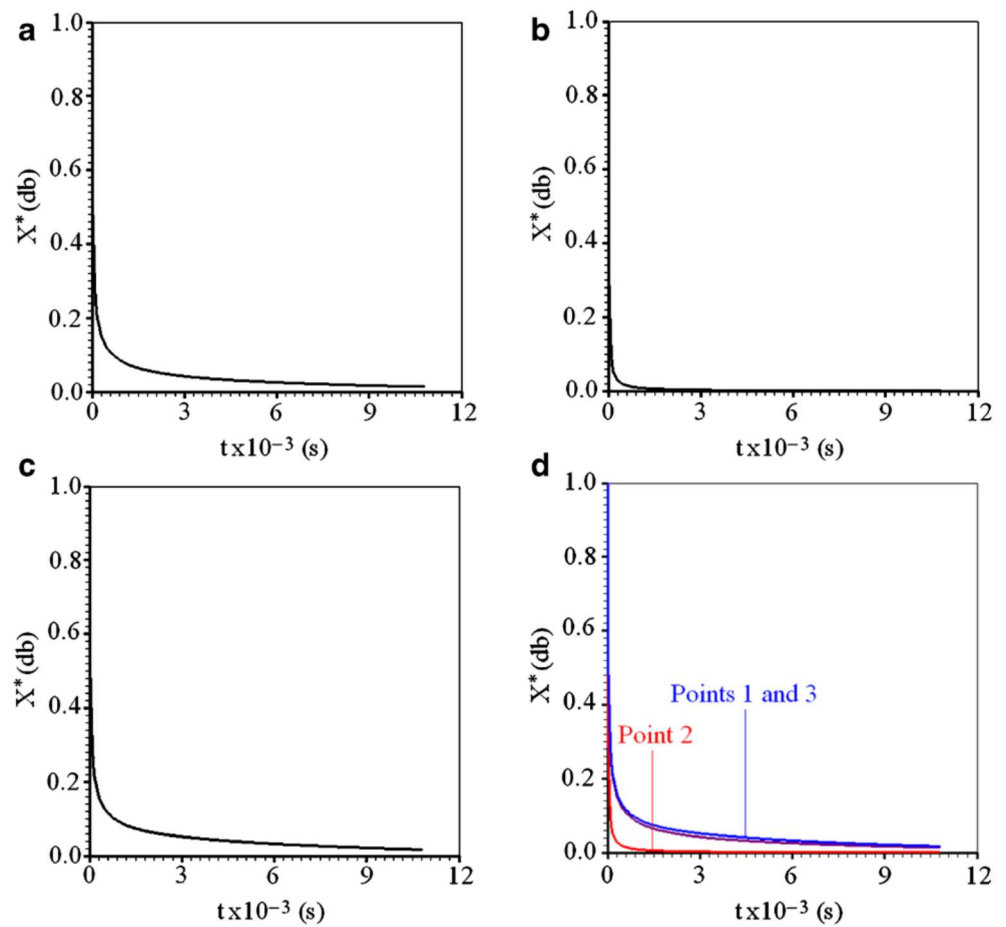


Fig. 7 Moisture distribution at instants: **a** $t=539.8$ s; **b** $t=2159.0$ s; **c** $t=3,779$ s; **d** $t=5,938$ s

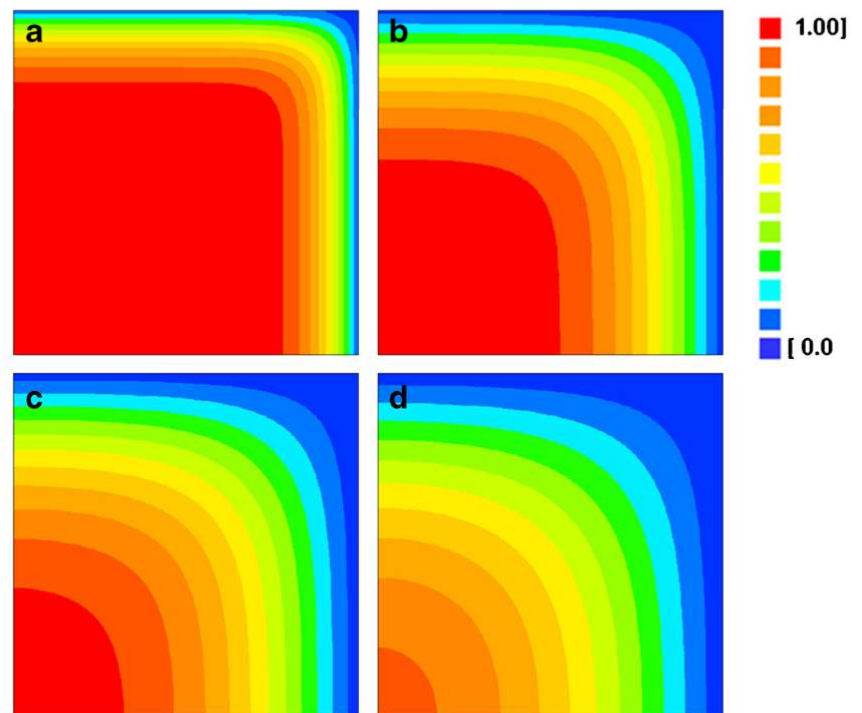


图 6 展示了在以下三个点处的局部水分含量随时间的变化情况: a 点、b 点、c 点; 以及曲线(a)、(b)、(c)的叠加情况。

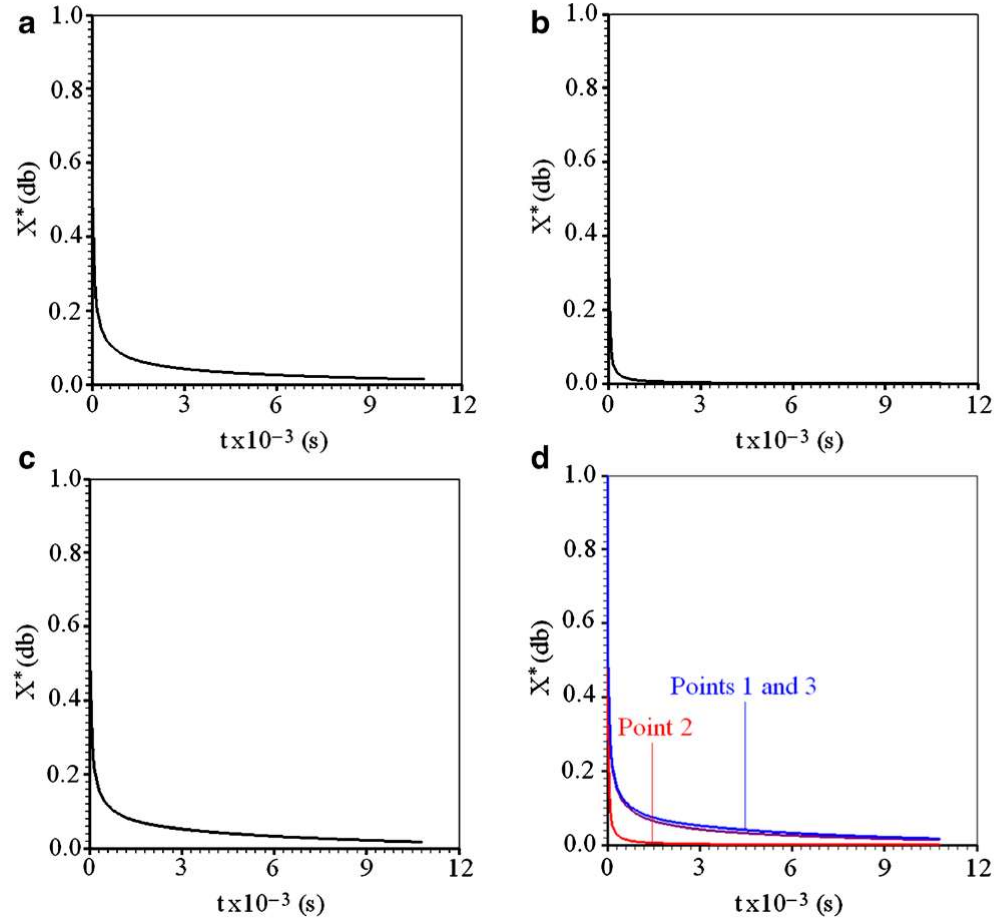
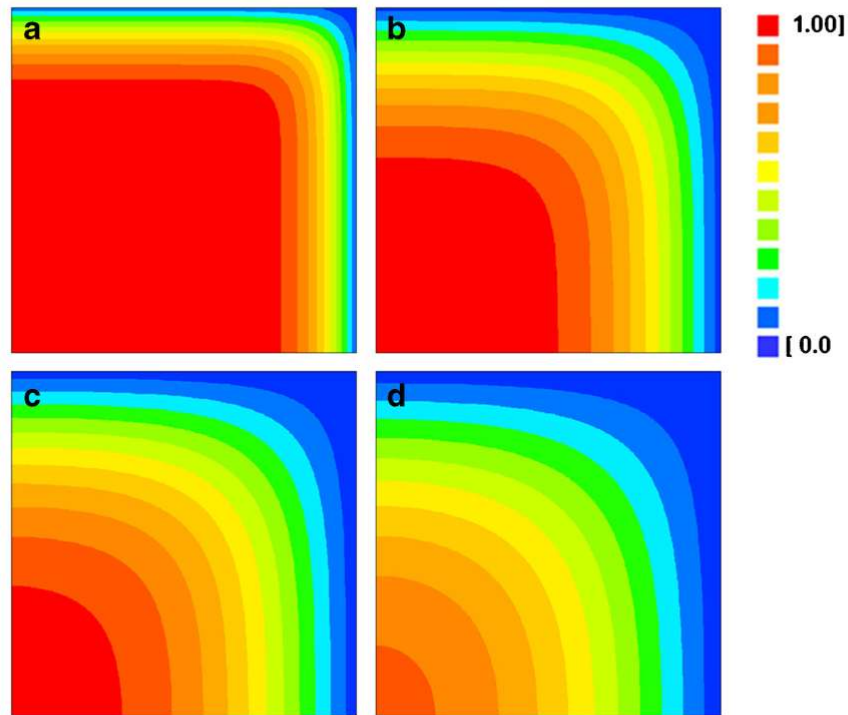


图 7 不同时间点的水分分布情况: a 为 t0539.8 秒时; b 为 t02159.0 秒时; c 为 t03,779 秒时; d 为 t05,938 秒时。



($\chi^2=3.5596 \times 10^{-4}$ and $R^2=0.9996$). In addition, the simulated curve has a distribution of errors that can be considered random, with an average value very close to zero.

The two models analyzed in this article presuppose restrictive assumptions as constant volume and effective mass diffusivity, and also homogeneous and isotropic medium. However, even if the obtained results were not considered completely satisfactory, these results could be used as initial values in other optimization processes which use numerical solutions. Such solutions would take into account the mentioned restrictions.

References

- Adaramoyea OA, Anjos RM, Almeida MM, Veras RC, Silvia DF, Oliveira FA, Cavalcante KV, Araújo IG, Oliveira AP, Medeiros IA (2009) Hypotensive and endothelium-independent vasorelaxant effects of methanolic extract from *Curcuma longa* L. in rats. *J Ethnopharmacol* 124:457–462
- Baronas R, Ivanauskas F, Juodeikiene I, Kajalavicius A (2001) Modelling of moisture movement in wood during outdoor storage. *Nonlinear Anal Model Contr* 6:3–14
- Bevington PR, Robinson DK (1992) Data reduction and error analysis for the physical sciences, 2nd edn. WCB/McGraw-Hill, Boston
- Chen JJ, Tsai CS, Hwang TL, Shieh PC, Chen JF, Sung PJ (2010) Sesquiterpenes from the rhizome of *Curcuma longa* with inhibitory activity on superoxide generation and elastase release by neutrophils. *Food Chem* 119:974–980
- Crank J (1992) The mathematics of diffusion. Clarendon, Oxford
- Da Silva WP, Silva CMDPS, Farias VSO, Gomes JP (2012) Diffusion models to describe the drying process of peeled bananas: optimization and simulation. *Dry Technol* 30(2):164–174
- Devahastin S, Niamnuy C (2010) Invited review: modelling quality changes of fruits and vegetables during drying: a review. *Int J Food Sci Tech* 45:1755–1767
- Hacihafizoglu O, Cihan A, Kahveci K, Lima AGB (2008) A liquid diffusion model for thin-layer drying of rough rice. *Eur Food Res Technol* 226:787–793
- Inano H, Onoda M (2002) Radioprotective action of curcumin extracted from *Curcuma longa* LINN: inhibitory effect on formation of urinary 8-hydroxy-2'-deoxyguanosine, tumorigenesis, but not mortality, induced by gamma-ray irradiation. *Int J Rad Oncol Biol Phys* 53:735–743
- Kim KI, Kim JW, Hong BS, Shin DH, Cho HY, Kim HK, Yang HC (2000) Antitumor, genotoxicity and anticlastogenic activities of polysaccharide from *Curcumazodoaria*. *Mol Cell* 10:392–408
- Kulasiri D, Woohead I (2005) On modelling the drying of porous materials: analytical solutions to coupled partial differential equations governing heat and moisture transfer. *Math Probl Eng* 2005:275–291
- Luikov AV (1968) Analytical heat diffusion theory. Academic Press, Inc. Ltd, London
- Mohanty I, Arya DS, Dinda A, Joshi S, Talwar KK, Gupta SK (2004) Protective effects of *Curcuma longa* on ischemia-reperfusion induced myocardial injuries and their mechanisms. *Life Sci* 75:1701–1711
- Mulet A, Blasco M, García-Reverter J, García-Pérez JV (2005) Drying kinetics of *Curcuma longa* rhizomes. *J Food Sci* 70:318–323
- Negi PS, Jayaprakasha GK, Rao LJM, Sakariah KK (1999) Antibacterial activity of turmeric oil: a byproduct from curcumin manufacture. *J Agric Food Chem* 47:4297–4300
- Olek W, Weres J (2007) Effects of the method of identification of the diffusion coefficient on accuracy of modeling bound water transfer in wood. *Transport Porous Med* 66:135–144
- Prasad JA (2009) Convective heat transfer in herb and spices during open sundrying. *Int J Food Sci Tech* 44:657–665
- Prasad J, Vijay VK, Tiwari GN, Sorayan VPS (2006) Study on performance evaluation of hybrid drier for turmeric (*Curcuma longa* L.) drying at village scale. *J Food Eng* 75:497–502
- Prathapan A, Likhman M, Arumughan C, Sundaresan A, Raghu KG (2009) Effect of heat treatment on curcuminoid, colour value and total polyphenols of fresh turmeric rhizome. *Int J Food Sci Tech* 44:1438–1444
- Quiles JL, Huertas JR, Battino M, Mataix J, Ramirez-Tortosa MC (2002) Antioxidant nutrients and adriamycin toxicity. *Toxicology* 180:79–95
- Ricard AP, Suárez JR, Berumen LA (2005) The drying of red oak at vacuum pressure. *Maderas Cien Tecnol* 7:23–26
- Silva WP, Precker JW, Silva CMDPS, Silva CDPS, Silva DDPS, Lima AGB (2007) Calculation of the uncertainty in the determination of the equilibrium moisture content of pumpkin seed flour. *Int J Food Eng* 3:1–10
- Silva WP, Precker JW, Silva CMDPS, Silva DDPS (2009) Determination of the effective diffusivity via minimization of the objective function by scanning: application to drying of cowpea. *J Food Eng* 95:298–304
- Silva WP, Precker JW, Silva CMDPS, Gomes JP (2010a) Determination of effective diffusivity and convective mass transfer coefficient for cylindrical solids via analytical solution and inverse method: application to the drying of rough rice. *J Food Eng* 98:302–308
- Silva WP, Silva LD, Silva CMDPS, Nascimento PL (2010b) Optimization and simulation of drying processes using diffusion models: application to wood drying using forced air at low temperature. *Wood Sci Technol* 45:787–800
- Silva CMDPS, Silva WP, Farias VSO, Gomes JP (2012) Effective diffusivity and convective mass transfer coefficient during the drying of bananas. *Eng Agr* 32:342–353
- Surh YJ (2002) Anti-tumor promoting potential of selected spice ingredients with antioxidative and anti-inflammatory activities: a short review. *Food Chem Toxicol* 40:1091–1097
- Taylor JR (1997) An introduction to error analysis, 2nd edn. University Science, Sausalito
- Wu B, Yang W, Jia C (2004) A three-dimensional numerical simulation of transient heat and mass transfer inside a single rice kernel during the drying process. *Biosystems Eng* 87:191–200
- Xia X, Cheng G, Pan Y, Xia ZH, Kong LD (2007) Behavioral, neurochemical and neuroendocrine effects of the ethanolic extract from *Curcuma longa* L. in the mouse forced swimming test. *J Ethnopharmacol* 110:356–363
- Yue GGL, Chan BCL, Hon PM, Kennelly EJ, Yeung SK, Cassileth BR, Fung KP, Leung PC, Lau CBS (2010) Immunostimulatory activities of polysaccharide extract isolated from *Curcuma longa*. *Int J Biol Macromol* 47:342–347

($\chi^2 0.5596 \times 10^4$ and $R^2 0.9996$)。此外, 模拟得到的曲线具有可以视为随机分布的误差分布, 其平均值非常接近零。

本文中分析的两个模型都假设了一些限制性条件, 比如体积恒定、质量扩散效应可以忽略不计, 以及介质是均匀且各向同性的。不过, 即使所获得的结果并不完全令人满意, 这些结果仍然可以作为其他采用数值解的优化过程的初始值。这样的数值解会考虑到上述限制条件。

参考文献

- Adaramoyea OA, Anjos RM, Almeida MM, Veras RC, Silvia DF, Oliveira FA, Cavalcante KV, Araújo IG, Oliveira AP, Medeiros IA (2009). 从长叶姜黄中提取的甲醇提取物对大鼠的降压作用及独立于内皮细胞的血管舒张作用。《民族药理学杂志》124 卷: 457–462 页。Baronas R, Ivanauskas F, Juodeikiene I, Kajalavicius A (2001). 关于木材在户外储存过程中水分迁移的建模研究。非线性分析模型控制 6:3–14 贝文顿 PR, 罗宾逊 DK (1992) 《物理科学的数据简化与误差分析》, 第 2 版。WCB/McGraw-Hill 出版社, 波士顿
- 陈 JJ, 蔡 CS, 黄 TL, 谢 PC, 陈 JF, 宋 PJ (2010) 来自长叶姜根茎中的三环烯类化合物, 具有抑制中性粒细胞产生超氧化物和释放弹性酶的作用。《食品化学》119:974–980 克兰克 J (1992) 扩散的数学原理。Clarendon 出版社, 牛津 达席尔瓦 WP, 席尔瓦 CMDPS, 法里亚斯 VSO, 戈梅斯 JP (2012) 用于描述去皮香蕉干燥过程的扩散模型: 优化与模拟。《干燥技术》30(2):164–174 德瓦哈斯蒂因 S, 尼安努伊 C (2010) 特邀评论: 水果和蔬菜在干燥过程中品质变化的建模: 综述。《国际食品科学技术杂志》第 45 卷: 1755–1767 页。Hacıhafızoglu O, Cihan A, Kahveci K, Lima AGB (2008). 糙米薄层干燥的液体扩散模型。《欧洲食品研究技术》第 226 卷: 787–793 页。Inano H, Onoda M (2002). 从 *Curcuma longa* 中提取的姜黄素的放射保护作用: 对 γ 射线照射引起的尿 8-羟基-2'-脱氧鸟苷形成有抑制作用, 但对肿瘤发生无影响, 对死亡率则无影响。《国际放射肿瘤学物理杂志》第 53 卷: 735–743 页。Kim KI, Kim JW, Hong BS, Shin DH, Cho HY, Kim HK, Yang HC (2000). Curcumazedoaria 多糖的抗肿瘤、遗传毒性及抗老化活性。《分子细胞》第 10 卷: 392–408 页。Kulasiri D, Woohead I (2005). 关于多孔材料干燥过程的建模: 求解控制热量和水分传递的耦合偏微分方程的分析解。Math Probl Eng 2005:275–291 卢伊科夫 AV (1968) 《分析性热扩散理论》。伦敦: Academic Press 公司。莫汉蒂 I, 阿里亚 DS, 丁达 A, 乔希 S, 塔尔瓦尔 KK, 古普塔 SK (2004) 《姜黄对缺血再灌注引起的心肌损伤的保护作用及其机制》。Life Sci 75:1701–1711
- Mulet A, Blasco M, García-Reverter J, García-Pérez JV (2005) 姜黄根茎的干燥动力学过程。J Food Sci 70:318–323
- Negi PS, Jayaprakasha GK, Rao LJM, Sakariah KK (1999) 姜黄油的抗菌活性: 作为姜黄素生产过程中的副产品。J Agric Food Chem 47:4297–4300
- Olek W, Weres J (2007) 扩散系数识别方法对木材中结合水转移建模准确性的影响。Transport Porous Med 66:135–144
- Prasad JA (2009) 在露天晾晒过程中, 草药和香辛料中的对流热传递现象。Int J Food Sci Tech 44:657–665
- Prasad J, Vijay VK, Tiwari GN, Sorayan VPS (2006) 关于乡村规模下姜黄干燥混合干燥机的性能评估研究。J Food Eng 75:497–502 普拉塔潘 A, 卢克曼 M, 阿鲁穆甘 C, 桑德拉斯 AN, 拉古 KG (2009) 热处理对新鲜姜黄根茎中的姜黄素类成分、颜色值及总多酚含量的影响。Int J Food Sci Tech 44:1438–1444 奎尔斯 JL, 韦尔塔斯 JR, 巴蒂诺 M, 马泰克斯 J, 拉米雷斯-托尔塔萨 MC (2002) 抗氧化营养素与阿德里霉素的毒性作用。Toxicology 180:79–95 里卡德兹 AP, 苏亚雷斯 JR, 贝鲁门 LA (2005) 在真空压力下红橡木的干燥过程。Maderas Cien Tecnol 7:23–26 席尔瓦 WP, 普雷克 JW, 席尔瓦 CMDPS, 席尔瓦 CDPS, 席尔瓦 DDPS, 利马 AGB (2007) 计算南瓜籽粉平衡水分含量测定中的不确定性。Int J Food Eng 3:1–10 席尔瓦 WP, 普雷克 JW, 席尔瓦 CMDPS, 席尔瓦 DDPS (2009) 通过扫描法最小化目标函数来测定有效扩散系数: 应用于豇豆的干燥过程。J Food Eng 95:298–304 Silva WP, Precker JW, Silva CMDPS, Gomes JP (2010a) 通过解析方法和反分析方法确定圆柱形固体的有效扩散系数和对流质量传递系数: 应用于糙米干燥过程。J Food Eng 98:302–308 Silva WP, Silva LD, Silva CMDPS, Nascimento PL (2010b) 优化方法
- 利用扩散模型对干燥过程进行建模与仿真: 应用于低温风干木材的干燥技术。木材科学技术 45:787–800
- Silva CMDPS, Silva WP, Farias VSO, Gomes JP (2012) 《香蕉干燥过程中有效扩散系数和对流传质系数研究》。Eng Agr 32:342–353
- Surh YJ (2002) 具有抗氧化和抗炎活性的几种香料成分的抗肿瘤潜力: 简要综述。Food Chem Toxicol 40:1091–1097
- Taylor JR (1997) 《误差分析导论》, 第 2 版。University Science, Sausalito
- Wu B, Yang W, Jia C (2004) 对单个米粒在干燥过程中瞬态热质和传质过程的三维数值模拟。Biosystems Eng 87:191–200
- Xia X, Cheng G, Pan Y, Xia ZH, Kong LD (2007) 从姜黄中提取的乙醇提取物在小鼠强迫游泳实验中的行为、神经化学和神经内分泌效应。J Ethnopharmacol 110:356–363
- Yue GGL, Chan BCL, Hon PM, Kennelly EJ, Yeung SK, Cassileth BR, Fung KP, Leung PCLau CBS (2010) 《从长叶姜黄中提取的多糖提取物的免疫刺激作用》。